

Impact of Task Switching and Work Interruptions on Software Development Processes

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ABSTRACT

Software developers often work on multiple projects and tasks throughout a work day, which may affect their productivity and quality of work. Knowing how working on several projects at a time affects productivity can improve cost and schedule estimations. It also can provide additional insights for better work scheduling and the development process. We want to achieve a better productivity without losing the benefits of work interruptions and multitasking for developers involved in the process. To understand how the development process can be improved, first, we identify work interruptions that mostly have a negative effect on productivity, second, we need to quantitatively evaluate impact of multitasking (task switching, work context switching) and work interruptions on productivity. In this research we study cross-project multitasking among the developers working on multiple projects in an educational setting. We propose a way to evaluate the number of cross-project interruptions among software developers using self-reported work logs. This paper describes the research that found: a) software developers involved in two or more projects on average spend 17% of their development effort on cross-project interruptions, b) the amount of effort spent on interruptions is overestimated by the G. Weinberg's heuristic, c) the correlation between the number of projects and effort spent by developers on cross-project interruptions is relatively weak, and d) there is strong correlation between the number of projects and the number of interruptions developers reported.

CCS CONCEPTS

• **Social and professional topics** → **Software management**; • **Software and its engineering** → **Programming teams**;

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KEYWORDS

Work interruptions, effort estimation, multitasking overhead, cost estimation, software development

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1 INTRODUCTION

Many have observed that software developers need to switch between different tasks quite often throughout a work day [1]. Parnin and Rugaber [1] studied 10,000 sessions of developers' work and concluded that work interruptions and task resumptions are a frequent problem for developers. Another study [7] found that information-technology workers spent an average of only 3 minutes per task before switching to another task. DeMarco and Lister [3] found that developers work only 30% of their time alone and spend another 70% on collaboration and interaction with each other. Often such collaboration introduces work interruptions and requires developers to work on several projects at a time. In general, working on multiple tasks at a time and switching between them can have both positive and negative effects.

In order to better estimate development costs and achieve a better productivity we want to know the negative impact of work interruptions. We chose to study cross-project interruptions, because this type of interruptions introduces a time-consuming context switching which is often viewed as a waste of time. To achieve a better productivity cross-project interruptions can be reduced in development processes (e.g. Lean, Agile processes).

In this research, we quantitatively evaluated the impact of cross-project context switching and work interruptions on cost overhead in several software development projects in an educational setting. We used observational data collected from developers' self-reported work logs.

We answer two research questions in this study: (1) whether there is a linear correlation between the number of cross-project work interruptions and the number of projects and (2) whether

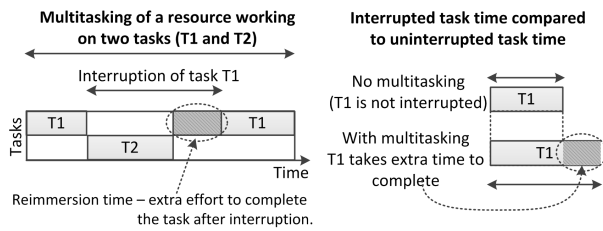


Figure 1: Reimmersion time.

there is a linear correlation between the amount of time spent on cross-project interruptions and the number of projects involved.

2 BACKGROUND

Dzubak [4] defines multitasking as the engagement in individual tasks that are performed sequentially and implies that there is necessarily some time spent switching between tasks. In other words, parallel tasks interrupt each other.

Jett and George [5] distinguish four major types of interruptions: intrusion, break, distraction, discrepancy. All of them have positive and negative effects on productivity, different resumptions time, different time of switching to a new work context, different underlining causes and sources of interruptions.

Positive effects of work interruptions in software development include a better team collaboration and communication (e.g. interruptions for design/peer review and architecture review meetings, which help developers to define a “big picture” of the project and obtain a feedback from their colleagues [4]), a stimulating work environment (e.g. reduced “the boredom effect” and the inclination to procrastinate [4]), a reduced “struggle time” trying to solve complex problems for too long [6]. Positive interruptions are typically related to software design work, problem solving, and debugging activities that often depend upon the creation and evolution of mental models. These mental models are what facilitate concurrent engineering and development.

Along with positive effects, studies [7-8] found that multitasking can reduce work productivity. In particular, interruptions introduce necessary time spent switching between the tasks, called “attention switching” in [8] and make an individual less productive by reducing the amount of short-term memory dedicated to a single task. Salvucci et al. [9] performed a set of experiments and observations to measure the time cost of task switching and determined that it depends on the complexity of the task, time length between task switches, and causes of interruptions (e.g. external or self-inflicted interruptions).

DeMarco and Lister [7] introduced a metric called reimmersion time as a measure of the effect of task interruption. Reimmersion time is an extra effort to complete the task after an interruption (Fig. 1). This metric is also known as resumption lag [1, 7, 9, 10]. G. Weinberg [11] introduced a heuristic measure of the effect of multitasking on reimmersion time that is a simple linear function of the number of projects assigned to an individual at the same time (see Fig. 6). The heuristic predicts a 100% increase of multitasking overhead for every additional task assigned.

In this paper, we study only cross-project interruptions and cross-project multitasking. We define cross-project multitasking as an involvement of software developers in multiple engineering activities with different or minimally overlapping contexts over a certain period. Cross-project interruptions are characterized by relatively long reimmersion times, longer resumption times, and significantly different work contexts of interrupting tasks [9]. Often such interruptions are external (not self-inflicted); therefore, could be limited by the development process (e.g. Agile and Lean development processes acknowledge the problem and introduce measures to reduce work interruptions [12, 13]).

Cross-project multitasking and interruptions may appear in different forms. A few examples are:

- Developers are often shared between projects in organizations with matrix structure
- Developers are shared between multiple releases within a single project (if several releases are maintained)
- In System of Systems environments, developers are often shared between several contexts such as customer-specific requirements, software releases for clients, etc.[13].

The focus of the rest of this paper is on negative productivity impacts of cross-project multitasking and interruptions.

3 CROSS-PROJECT MULTITASKING AND INTERRUPTIONS

3.1 Research Questions

In this paper, we measure a negative impact of cross-project interruptions as the total effort (person-hours) each developer spends on reimmersion time after each interruption that was caused by tasks from other activities.

The following research questions are addressed in the analysis below: is there a linear correlation between the number of projects each developer was involved with and

- a the number of cross-project interruptions they experience?
- b the amount of effort spent on cross-project interruptions?

There were two types of hypotheses tested for each question:

- H.1 Correlation between subject means: developers working on more projects tend to experience more cross-project (a) interruptions, (b) spent more effort.
- H.2 Correlation within subjects: an increase of the number of projects per subject is associated with an (a) increase of the number of cross-project interruptions, (b) increase of time spent on interruptions.

Additionally, we also compared the effort evaluation results with the predictions of G. Weinberg’s heuristic (Fig. 6).

3.2 Data Collection

In order to evaluate the impact of cross-project multitasking, we observed 10 software projects for three months. All projects were developed as part of a graduate level software engineering class at the University of Southern California. Clients of these projects represent various non-profit organizations, entrepreneurs and private companies. Clients joined teams of students of software engineering class to work on their projects. Most of the projects were scoped

to fit into a single semester of the class. Working on real life projects and reporting project progress is part of the class curriculum.

In total, we collected work logs and weekly progress reports of 68 students for one semester of work. All students had degrees in computer science related fields and experience in software development.

Daily work logs were part of the work process, which allowed us to collect data on development effort and acquire development effort distribution across projects and individuals without interfering with their work flow. Project tasks were assigned to developers via the project tracking system used in the class. Every student reported their daily and weekly progress (time spent) on each task, that he/she performed on that day. In addition to the work logs, students also provided weekly information about work interruptions and self-evaluated reimmersion time. They were instructed how to evaluate their reimmersion time and how to use the project tracking system.

Some of the students also took one or two other classes and had no outside jobs; therefore, they had to distribute their time between the following three types of activities: (1) a project in the software engineering class, (2) group projects/tasks/assignments in classes other than the software engineering class, (3) individual assignments in the software engineering class and all other classes. These three types of activities have very different context of work, and independent deadlines. We viewed a switching between these three types of activities as a cross-project interruption.

In this study, we collected detailed information about projects in the software engineering class via the project tracking system (Atlassian Jira). The project tracking system tracked tasks' statuses. A task's status, reported by developers, can show if it is completed, in progress or in the backlog. A task's labels also allowed us to distinguish different types of engineering activities (requirement engineering, coding, testing, documentation, etc.).

We collected information about the impact on the class project from the other two types of activities via weekly progress reports, where students provided self-evaluated reimmersion time and types of activities they were involved with throughout the week.

For the purpose of consistent terminology in this paper, we will call the three types of activities listed above as projects (activities with different context of work). We also will count switching between them as cross-project switching.

In order to analyze the impact of interruptions on effort, we selected a subset of 29 full-time students who worked in similar conditions. They only took two classes (one of them was the software engineering class) and had no outside jobs. At least 25% of their time in the class project was dedicated to development activities (programming, bug fixing, etc.).

3.3 Calculation of Work Interruptions and Reimmersion Time

We used reimmersion time as described in Fig. 1 to evaluate the impact of cross-project interruptions on developers' productivity. In order to calculate the total effort each developer spends on reimmersion, we need to know the number of interruptions (obtained from work logs) and the reimmersion time after each interruption (evaluated by developers).

Work interruptions were counted by analyzing developers' work logs and weekly progress reports. These two sources of information were used to verify the information about the number of projects students were involved with and the number of interruptions they experienced. Weekly progress reports provide the number of interruptions evaluated by developers, and work logs allow us to see how developers worked on a software engineering class project throughout the week. Since all 29 developers spent their work time only on up to three projects at a time (including the software engineering class project), their work logs of tasks from the software engineering class projects showed when work was interrupted. In some cases work logs can only provide a lower bound estimate on the number of interruptions, because we were only able to track one out of three projects they worked on.

Reimmersion time may depend on many factors: the complexity of the task, length of the interruption, complexity of the other task that interrupted work, individual's personality (closure-oriented vs. curiosity-oriented), etc.[5]. To account for all the factors that impact the reimmersion time we asked developers to evaluate their weekly average reimmersion time for tasks that they did for the class project. The total weekly effort spent on cross-project interruptions (i.e. multitasking overhead) is computed as *the total number of interruptions times the reimmersion time*.

4 RESULTS

In our observational data we collected repeated measures for the same set of projects over the course of 12 weeks (12 weeks = 12 times); therefore, we had to account for repeated observations in our linear regression analysis. For both effort and interruptions we tested two types of hypotheses: (1) developers working on more projects tend to have more interruptions and tend to spend more effort on multitasking overhead; (2) the increase of the number of projects worked on is associated with increase of the number of interruptions and effort.

To answer the first group of questions we used the linear correlation between subject means (mean values of the number of projects each developer was involved in, effort and the number of cross-project interruptions of each developer). Fig. 2 shows the correlation between the average number of interruptions and the average number of projects per week per developer. The total number of data points is 29. The linear correlation is relatively strong ($R^2 = 0.60699$). It supports the hypothesis H1.a that developers working on more projects tend to experience more interruptions. Fig. 3 shows the correlation between the average effort spent on context switching (cross-project multitasking overhead) and the average number of projects per week per developer. The total number of data points is also 29. The linear correlation is weak ($R^2 = 0.1622$). It does not support the hypothesis H1.b that developers working on more projects tend to have more cross-project multitasking overhead. From Fig. 6 we can see that average multitasking overhead remains the same.

To answer the second group of questions we used a multiple linear regression analysis to compute correlation coefficients. We treated developers as a categorical factor (a predictor). Fig. 4 shows the number of interruptions versus the number of projects per week. The total number of data points is 348 (repeated observations

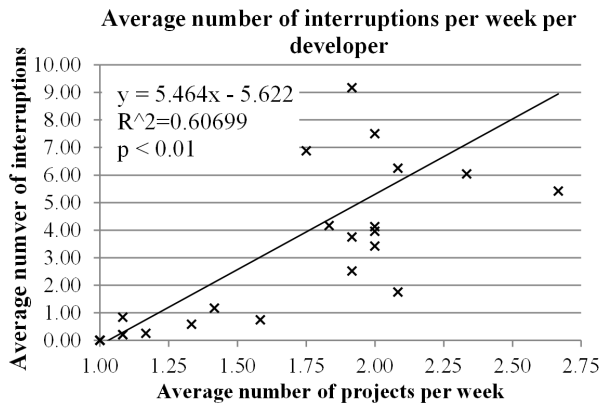


Figure 2: The average number of interruptions vs. the average number of projects per week per developer.

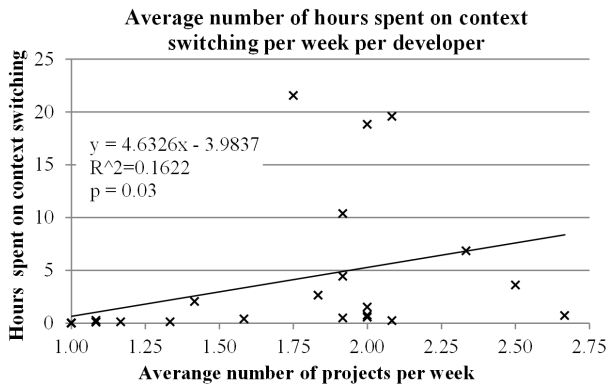


Figure 3: The average effort (person hours) vs. the average number of projects per week per developer.

included). The linear correlation is strong (multiple $R^2 = 0.7779$). It supports the hypothesis H2.a. This means that when a developer works on three projects in one week, he/she experiences more interruptions compared to weeks when he/she works on two projects. If developers work on one project, there is no cross-project interruptions at all. Fig. 5 shows the effort spent on context switching versus the number of projects per week. The total number of data points is 348 (repeated observations included). The linear correlation is strong (multiple $R^2 = 0.6934$). It supports the hypothesis H2.b. It means that when a developer works on three projects in one week, he/she has more multitasking overhead compared to weeks when he/she works on two projects.

To better see how multitasking overhead affects productivity, we converted effort spent on cross-project interruptions (measured in person hours) into the percentage of the full-time equivalent (FTE). FTE = 40 person hours per week.

Fig. 6 shows average effort each developer spent on cross-project multitasking overhead per week. It also compares multitasking overhead results from observations with the G. Weinberg's heuristic, which predicts a larger multitasking overhead.

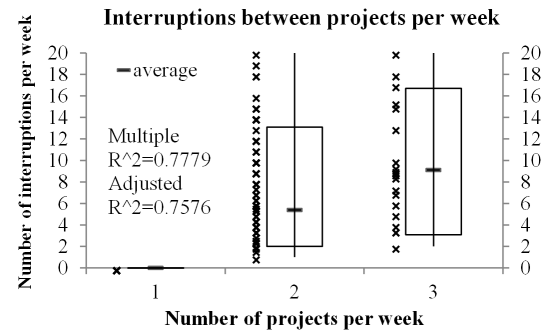


Figure 4: The number of cross project interruptions vs. the number of projects per week per developer.

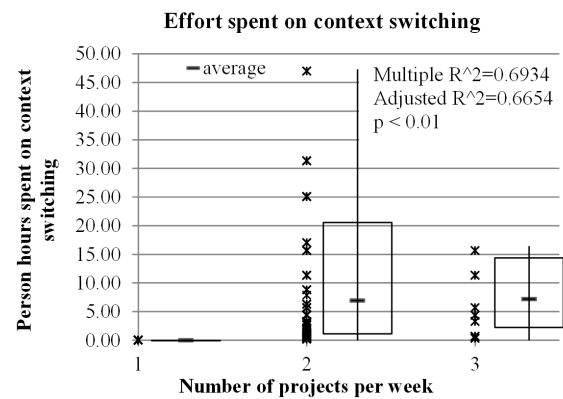


Figure 5: Effort spent on context switching vs. the number of projects per week per developer.

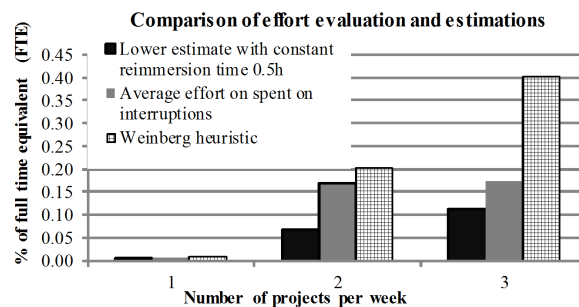


Figure 6: Average effort developers spent on cross-projects multitasking per week as a portion of FTE.

Different sources estimate that the reimmersion time for information/knowledge workers varies from 20 minutes to 1-2 hours [1, 2, 7, 9]. We evaluated a lower bound estimate for the multitasking overhead using a constant 0.5 hour value of the reimmersion time for all interruptions (see Fig. 6). The evaluated lower bound underestimates the actual average value of the multitasking overhead.

Additionally, we wanted to see if there was any impact of cross-project multitasking on quality of work. Students were evaluated during architecture reviews meetings and by the clients at the end

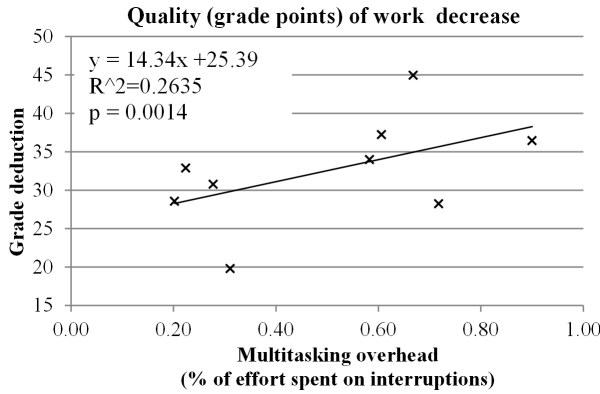


Figure 7: Quality of work decrease depending on multitasking overhead.

Table 1: Hypotheses summary.

Hypothesis	R^2 / multiple R^2	p	Accepted / Rejected
H1.a	0.60699	<0.01	Accepted
H2.a	0.7779	<0.01	Accepted
H1.b	0.01622	<0.03	Rejected
H2.b	0.6934	<0.01	Accepted

of the semester. The impact on the quality of work was measured in terms of grade points deducted. Fig. 7 shows grade deduction depending on the amount of multitasking overhead.

In order to make quantitative conclusions about correlation between the quality of work and the amount of multitasking overhead, we need to collect more data. However, visual observation of the data points suggests that projects where multitasking overhead took less than 40% of the overall effort had a better quality of work compare to other projects. Table 1 summarizes all other results.

5 TREATS TO VALIDITY

This is an on-going research which has limitations discussed below.

Self-tracking of the reimmersion time and hours devoted to projects may not be accurate. However, these values do not affect our conclusions about the correlation between the number of projects and the number of interruptions. To get a more accurate evaluation of the reimmersion time, we plan to conduct a more controlled experiment.

Counting only cross-project interruptions might underestimate the overall multitasking overhead caused by internal (within project) and self-inflicted interruptions. Therefore, evaluations presented here can be viewed as a lower bound of the overall multitasking overhead.

Participating subjects were graduate students in software engineering educational environment. To limit the impact of deadlines in other classes, students in the selected sample were enrolled in the same classes (e.g. students who took 3 classes were all enrolled in software engineering class, algorithms class and AI class). The

method used in this research can be applied to industry projects in other domains.

6 CONCLUSIONS

In this study, we evaluated the impact of cross-project multitasking on software development effort by analyzing work logs and weekly progress reports. Results show that developers who work on 2 or 3 projects spend on average 17% of their effort on context switching between tasks from different projects. Developers who were involved in more projects tend to have more cross-project work interruptions. However, the linear correlation between the number of projects each developer worked on in one week and the amount of effort developers spend on context switching is weak. This partially can be explained by excessive amount of switching between two most active projects.

The number of interruptions can be used successfully to evaluate the quantitative impact of multitasking on effort. The number of interruptions that people experience can also be used as a metric and a measure of multitasking in teams. Additionally, the impact of multitasking on effort can be integrated into software development parametric cost and schedule estimation models such as the Constructive Cost Estimation Model (COCOMO©) for better effort estimation.

Finally, the work log analysis tools that we used in our research can be integrated with project tracking systems such as Atlassian Jira to provide real-time information about work interruptions and their impact on productivity.

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